Kitchnos — Business Requirements Document (BRD)

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Field	Value
Product	Kitchnos (working name: Kitchen AI)
Company	Drishix
Document	Business Requirements Document, v1.0
Status	Draft for approval
Companion documents	Kitchnos PRD · Architecture CSV pack · GitHub project setup

1. Executive summary

Kitchnos is an AI kitchen companion for Indian households. It knows what is in the kitchen, decides what to cook, guides cooking step by step by voice in the user's own language, and learns from every meal. The product's defensible asset is the **Kitchen Brain**: one persistent context of pantry, household, taste memory and cooking history that every feature reads from and writes back to.

The business case rests on three facts. Deciding what to cook is a daily, stressful job (about 37 hours a year per adult). Indian home diets are measurably unbalanced (unhealthy diets drive 56.4% of India's disease burden). And no product combines pantry awareness, Indian cuisine depth and native-language voice guidance.

This BRD asks for approval to:

1. Build V1, the Personal Kitchen Brain with Voice Cook inside Cook mode, over six two-week sprints (proposed start 16 Nov 2026).
2. Run a four-week closed beta with two segments, urban home cooks and voice-first rural cooks, before public launch.
3. Hold V1.5 and V2 (scanning, grocery agent, paid tier) behind the beta success gates in section 3.

2. Business problem and opportunity

Indian households cook daily but decide, shop and learn badly, and today's apps solve only one slice of that. The opportunity is a single companion that removes the daily decision, prevents waste and teaches skill in the user's own language.

Problem	Evidence	Business implication
Deciding what to cook is a daily tax	Adults spend about 37 hours a year choosing meals; 47% have eaten something unwanted to avoid deciding (Dolmio/OnePoll)	A daily-use hook: "Tonight, decided"
Indian cooks feel it most in the morning rush	80% of Indian moms dislike breakfast prep; nearly 30% call the menu decision their biggest challenge (V-Guard/Momspresso)	High frequency: two to three decisions a day
Home diets are unbalanced	Unhealthy diets cause 56.4% of India's disease burden (ICMR-NIN 2024); cereals, oil and sugar give over two-thirds of household energy (CEEW)	Health nudges inside familiar dishes
Food is wasted at home	About 55 kg per person per year, roughly 78 million tonnes (UNEP Food Waste Index 2024)	"Use Soon" saves money users can feel
Recipes don't teach judgment	Indian cooking runs on *andaza* (sight, smell, sound), learned by watching (Spice Spoon)	Voice and photo checkpoints become the teacher
Many users can't use text-heavy apps	51.6% of rural women don't own a phone but 76.3% use one, mostly shared (NSO CMS-T 2025)	Voice-first, no-login use on shared phones
Restocking is already instant	Quick commerce runs over 9 million orders a day (NewsBytes)	Grocery agent rides existing behaviour

Why existing products fall short. Recipe libraries became a commodity: Whirlpool laid off Yummly's whole team in April 2024 (The Spoon). Samsung Food ties its best features to Samsung fridges and a Western catalogue (Engadget). Pantry trackers drift out of sync because upkeep never ends (Plan to Eat). None speaks Indian languages with Indian kitchen judgment.

Enabling shift. Indian-language speech models are now good and cheap enough. Sarvam's speech-to-text covers 22 Indian languages plus code-mixed Hinglish at ₹0.49 per minute (Sarvam), which makes a voice cook affordable at scale.

3. Business objectives and success metrics

V1 succeeds if households cook with Kitchnos at least three times a week and the voice cook works for people who cannot read the app. Targets are proposals for the closed beta; they become commitments after the first two weeks of beta data.

ID	Objective	KPI	Target (beta, proposed)	Gate for
OBJ-1	Own the daily cooking decision	"Tonight, decided" acceptance rate	⩾ 40% of shown picks accepted	V1 public launch
OBJ-2	Build a cooking habit	Cooking sessions per active household per week	⩾ 3	V1 public launch
OBJ-3	Deliver the first magic moment fast	New users who finish a guided meal within 7 days	⩾ 40%	V1 public launch
OBJ-4	Make voice work for everyone	Cook sessions finished hands-free (no taps after start)	⩾ 60%	Rural rollout
OBJ-5	Feel instant	Voice reply latency (end of speech to first audio)	p50 ⩽ 1.0 s, p95 ⩽ 1.8 s	Rural rollout
OBJ-6	Keep the pantry trustworthy	Inventory corrections per household per week	⩽ 5, falling over time	V1.5 scanning
OBJ-7	Cut household waste	Use Soon items cooked before expiry	⩾ 50%	V2
OBJ-8	Retain users	Day-30 household retention	⩾ 25%	Paid tier (V2)
OBJ-9	Stay affordable at scale	Variable cost per cooking session	⩽ ₹5	Free tier sizing
OBJ-10	Earn love	Net Promoter Score	⩾ 50	Public launch marketing

ID	Objective	KPI	Target (beta, proposed)	Gate for
OBJ-11	Monetise (V2)	Free-to-paid conversion of monthly active households	⩾ 4%	V3 investment

Success for the rural segment is measured by observation as well as data: a first-time user finishes a dish without anyone helping her use the phone.

4. Target customers and segments

Launch with two segments that stress the product in opposite ways: urban home cooks test intelligence and polish, rural voice-first cooks test simplicity. The other segments follow once the Kitchen Brain is proven.

Segment	Who	Main job to be done	Language and device	Release	Revenue role
S1 Family home cooks (launch)	Cooks for 3–6 people, urban or semi-urban, often a homemaker	Decide 2–3 meals a day that suit everyone, with what's at home	Hindi or regional + Hinglish; mid-range Android	V1	Core paying household
S2 Voice-first rural cooks (launch, pilot)	Women in small towns and villages, often low literacy, shared family phone	Cook favourite dishes and variations from what's available, by talking	Hindi dialects and regional languages; entry Android, patchy data	V1 pilot	Sponsored or institution-funded
S3 Urban working professionals	22–35, cook for themselves or a partner, time-poor	Quick, healthy meals; less waste; less ordering in	English or Hinglish; iPhone or premium Android	V1	Highest willingness to pay
S4 Beginners and people living away from home	Students, first jobs, newly married	Learn to cook reliably; recreate family dishes	Hinglish	V1	Free-to-paid via skill features
S5 Households with a domestic cook	Planner and cook are different people	Plan menus, hand them to the cook in her language	Planner: English; cook: regional voice	V2	Premium household plan

Segment	Who	Main job to be done	Language and device	Release	Revenue role
S6 Partners (B2B2C)	Grocery platforms, FMCG and appliance brands, NGOs, CSR programmes	Reach engaged home cooks; fund nutrition access	—	V2 onward	Commissions, sponsorships, licences

5. Scope

V1 delivers the full loop from your blueprint's first milestone (kitchen → craving → recommendation → cook → feedback → memory → inventory update), with voice inside Cook mode. Everything that depends on partners or heavy data comes later.

Capability	V1	V1.5	V2	V3
Living pantry: Fridge, Pantry, Freezer, Staples, homemade components	In			
Natural-language and voice pantry entry	In			
Barcode, camera and receipt scanning		In		
"Tonight, decided" daily pick + craving search	In			
Household profiles (spice, diet, "family plates")	Basic (up to 6 members)		Full	
Curated Indian recipe base (300 dishes, launch region + North Indian)	In	500	1,000+	
Recipe adaptation (substitutes, equipment, servings, spice)	In			
Cook With Me with timers and photo checkpoints	In			
Voice Cook in Cook mode (Hindi, Hinglish, English + 1 regional language)	In	+3 languages	+6 languages	All 22

Capability	V1	V1.5	V2	V3
Voice on Home and Kitchen (craving, pantry)	In			
Recipe Rescue	In			
Feedback, taste memory with Edit · Keep · Forget	In		Advanced	
Use Soon and leftovers	Use Soon		Leftovers	
Shopping list (missing items, running low)	In			
Weekly plate health nudges		In		
Grocery agent with official provider integrations			In	
Predictive restocking			In	Auto (approved)
WhatsApp voice-note channel			In	
Paid tier (Kitchnos Plus)			In	
Cook handoff to domestic cook			In	
Phone-call (IVR) channel, family recipe capture				In

Out of scope for all releases: medical diet prescriptions or diagnosis; silent purchases without user approval; scraping or unofficial grocery integrations; restaurant ordering; storing raw voice recordings by default.

6. Business requirements

Twenty requirements define what the business needs from the product; the PRD turns each into functional requirements (the "PRD area" column) and the GitHub backlog traces every user story back to a BR.

ID	Requirement	Rationale	Priority	Release	PRD area
BR-01	Keep one persistent Kitchen Brain per household that every feature reads and writes	The moat is shared context, not single features	Must	V1	KIT, LRN
BR-02	Suggest one dinner pick a day from pantry, taste, household and recent meals, with a one-line reason	Owns the daily decision; drives habit	Must	V1	HOME
BR-03	Let users add pantry items by voice and natural language in under 10 seconds per haul	Manual upkeep kills pantry apps	Must	V1	KIT, INP
BR-04	Update inventory automatically after cooking, with easy correction	Pantry must stay true without effort	Must	V1	KIT, COOK
BR-05	Guide cooking step by step with timers, photo checkpoints and sensory cues	Teaches *andaza*; cuts failure	Must	V1	COOK
BR-06	Provide a real-time voice cook in Hindi, Hinglish, English and one regional language at launch	Reaches low-literacy users; hands are busy while cooking	Must	V1	VOICE
BR-07	Answer voice turns in about one second	Slow voice feels broken and is abandoned	Must	V1	VOICE
BR-08	Adapt any recipe to missing ingredients, equipment, servings and spice	"Cook from what you have" is the promise	Must	V1	REC
BR-09	Support multiple household members with different spice and diet needs	Indian kitchens cook one pot for many people	Must	V1	HH
BR-10	Learn from explicit feedback and behaviour, and let users edit, keep or forget each memory	Personalisation plus trust and control	Must	V1	LRN
BR-11	Rescue common cooking failures in the moment	One failure often ends the habit	Must	V1	RES
BR-12	Turn food nearing expiry into meal suggestions	Saves money; visible value	Should	V1	LEFT
BR-13	Keep cooking working offline once a session starts	Patchy networks in target regions	Must	V1	COOK, VOICE
BR-14	Work without email or password on shared phones	Rural women mostly share phones	Must	V1	ONB, PRIV

ID	Requirement	Rationale	Priority	Release	PRD area
BR-15	Comply with India's Digital Personal Data Protection Act 2023, with explicit consent for voice and memory	Legal requirement; trust	Must	V1	PRIV
BR-16	Give weekly, non-numeric health nudges aligned with ICMR-NIN guidance	Health without calorie tracking	Should	V1.5	HLTH
BR-17	Build a shopping list from missing and running-low items	Bridges cooking and buying	Must	V1	SHOP
BR-18	Order groceries only through official provider integrations and only after explicit approval of items and total	Revenue path without risking trust or legality	Should	V2	GROC
BR-19	Keep variable cost per cooking session at or below ₹5	Makes a free tier and rural access viable	Must	V1	NFR
BR-20	Instrument every journey for the section 3 KPIs	Beta gates need data	Must	V1	ANL

7. Business model and pricing

Kitchnos is free to start and earns from a household subscription, grocery commissions and partner programmes. Prices below are proposals to test in beta, not decisions.

Plan	Price (proposed)	Includes
Free	₹0	Pantry, daily pick, recipes, Cook With Me (touch), 4 voice-cook sessions a month, basic memory, shopping list
Kitchnos Plus	₹149 a month or ₹999 a year	Unlimited voice cook, up to 6 household profiles, weekly plate, advanced memory, leftovers, priority grocery agent
Community access	Funded by partner	Voice-first access for rural households, paid per active household by an NGO, CSR or government nutrition programme

Revenue streams, in order of timing:

1. **Subscriptions** (V2): Kitchnos Plus, sold in-app. Google Play takes a 15% fee on subscriptions.

2. **Grocery commissions** (V2): only through official partner integrations; rates to be negotiated per provider.
3. **Brand and appliance partnerships** (V2): clearly labelled sponsored recipes and appliance-specific guided cooking. A sponsored ingredient is never forced into a recipe and never hidden.
4. **Institutional licences** (V2 pilot): CSR and nutrition programmes fund the rural voice cook; the pilot doubles as impact evidence.

An in-house brand such as BilonaPure ghee can serve as the first sponsored-recipe pilot to test the partnership format before external brands.

8. Unit economics

A 15-minute voice-guided cooking session costs about ₹5.2 with the recommended architecture, against about ₹43 if built naively. Four design choices make the difference, and they are written into the PRD as requirements.

Cost per 15-minute voice cook session (₹, at ₹88 per US$)

Cost line	Naive build	Recommended build	What changes
Speech-to-text (Sarvam, ₹0.49/min pay-as-you-go, ₹0.41 Growth)	7.35	1.64	Voice detection in the agent sends only her ~4 min of speech to recognition, not 15 min of kitchen noise
Text-to-speech (Sarvam Bulbul V4 Flash, ₹20 per 10K characters, ₹17 Growth)	6.00	1.36	Step audio pre-generated once per recipe, language and voice, then cached; only live answers (~800 characters) synthesised
Language model	16.37	1.06	Claude Haiku 5.5 for conversation turns with prompt caching; Claude Sonnet 5.5 only to plan new or heavily adapted dishes (assumed 1 in 4 sessions)

Cost line	Naive build	Recommended build	What changes
Real-time transport (LiveKit)	13.20	0.88	Self-hosted agent worker billed as WebRTC minutes ($0.0005/min) instead of hosted agent minutes ($0.01/min); disconnect during long timers
Agent compute	—	0.30	Assumption: shared CPU workers
Total	**42.92**	**5.24**	

A touch-only session (no voice) costs about ₹1, mostly language-model calls.

Per household per month (assumptions: Plus household cooks 12 times, 60% by voice; free household cooks 6 times, 4 by voice):

Line	Plus household	Free household
Cooking sessions	₹42.5	₹23.0
Daily picks, scans, infrastructure (estimate)	₹9.6	₹6.0
Variable cost	**₹52**	**₹29**
Net revenue after 15% Play fee	₹126.65	₹0
Contribution	**≈ ₹75 (59%)**	**≈ −₹29**

Decision needed before beta. At 4% conversion, subscriptions alone cannot fund a free tier that costs ₹29 a month per household. Options: cap free voice to 2 sessions; fund free and rural voice through grocery commissions and partner programmes; or both. The beta should measure real session length and talk time, since every number above is an estimate.

Prices: Sarvam API pricing; LiveKit Cloud plans; Claude Haiku 5.5 at $0.10 / $0.50 and Sonnet 5.5 at $2 / $10 per million input / output tokens, from third-party summaries (Digital Applied, eesel), to confirm on Anthropic's pricing page. Sarvam's Bulbul discount ends 12 Nov 2026, so list prices are used.

9. Go-to-market plan

Win one region deeply before going wide: Rajasthan and North Indian home cooking first, Android first, Hindi and Hinglish first.

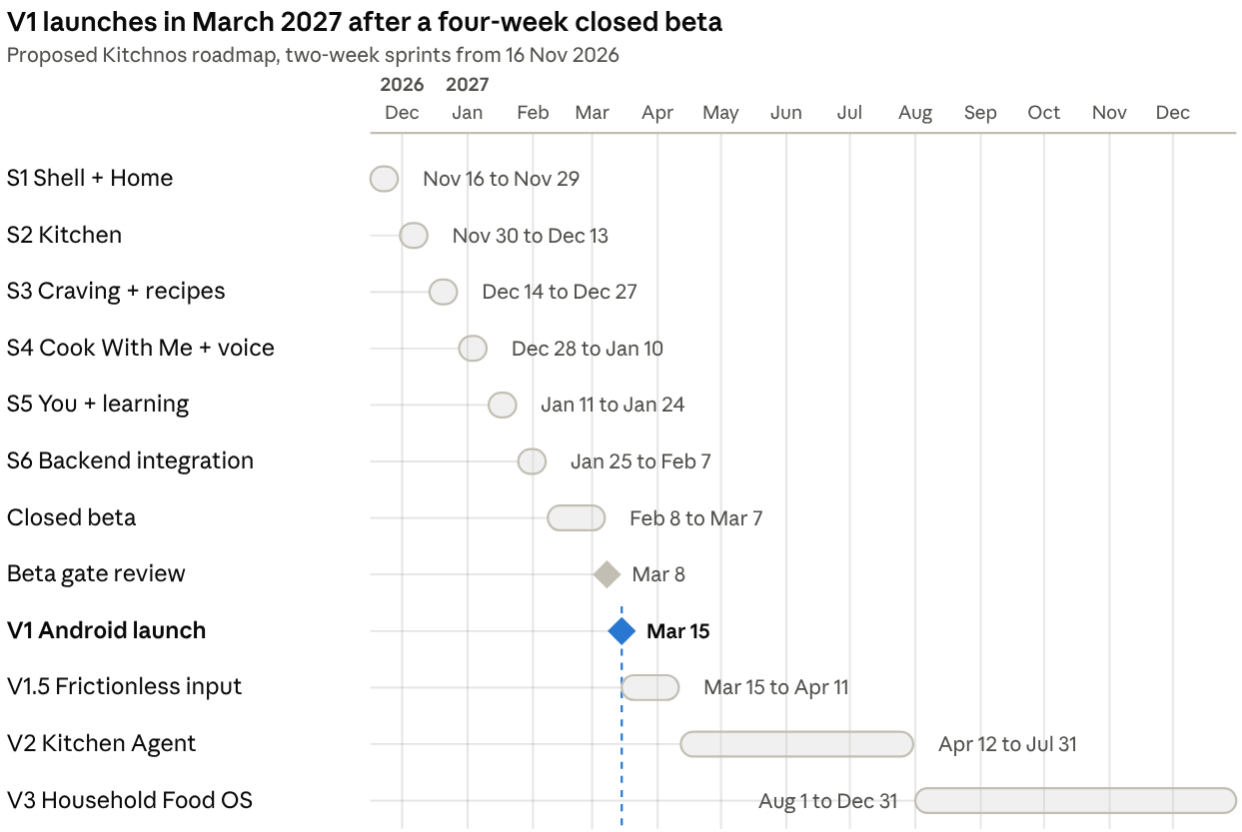
Phase	Audience	Channels	Proof point to move on
1. Closed beta (4 weeks)	150 urban and semi-urban households (S1, S3) + 30 rural cooks (S2)	Personal networks, cooking WhatsApp groups, one NGO or self-help-group (SHG) partner for the rural pilot	Section 3 beta gates met
2. Android launch, one region	S1, S3, S4 in Hindi-speaking North India	Short-form cooking reels showing "what can I make with this" on real fridges; creator partnerships; Drishix's own performance-marketing stack	Day-30 retention ≥ 25%, cost per activated household known
3. Languages and iOS	Add 3 regional languages; iOS	Regional creators; referrals ("cook this together" share)	Each language reaches the same activation as Hindi
4. Monetise and partner	All segments; S5 household plans	Kitchnos Plus, grocery partners, brand pilots, CSR licences for rural access	Conversion ≥ 4%; first partner revenue

Positioning line: "It knows my kitchen and my family, decides tonight's dinner, and makes me a better cook every time."

Growth loops to build in: a shareable "made with Kitchnos" dish card; family members joining a household; family recipe capture (V3) shared to relatives.

10. Roadmap and milestones

Six two-week sprints build V1, following the blueprint's sprint plan, with voice added to Cook With Me in sprint 4. A four-week closed beta and a gate review come before the Android launch; V1.5, V2 and V3 follow the blueprint roadmap.



proposed roadmap · Nov 2026 to Dec 2027

The gate review checks the section 3 targets; if they are missed, launch slips and the beta extends rather than scope growing. Backend services are built in parallel from sprint 1; sprint 6 swaps the app's mocks for live services, as the blueprint plans.

11. Stakeholders and responsibilities

The blueprint's team roles stay; V1 adds three roles it needs but did not name: a voice/AI engineer, a culinary lead and a field research partner. R = responsible, A = accountable, C = consulted, I = informed.

Activity	Product owner	Frontend dev (+ Antigravity)	Backend / AI (Claude)	Voice / AI engineer	Design (Figma, Stitch)	Culinary lead	QA	Field partner (NGO/SHG)	Privacy advisor
Requirements and priorities	A	C	C	C	C	C	I	C	C
Visual design and design system	A	C	I	I	R	I	I	I	I
Mobile app (React Native + Expo)	A	R	C	C	C	I	C	I	I

Activity	Product owner	Frontend dev (+ Antigravity)	Backend / AI (Claude)	Voice / AI engineer	Design (Figma, Stitch)	Culinary lead	QA	Field partner (NGO/SHG)	Privacy advisor
Supabase schema, services, Kitchen Brain	A	C	R	C	I	C	C	I	C
Voice agent and speech pipeline	A	C	C	R	I	C	C	C	C
Recipe base, variation rules, *andaza* cues	A	I	C	I	I	R	C	C	I
Testing on real devices and kitchens	A	C	C	C	I	C	R	C	I
Rural pilot and field observation	A	I	I	C	C	C	C	R	C
Consent, data protection (DPDP Act)	A	C	R	C	I	I	I	I	R
Partnerships and pricing	R/A	I	I	I	I	C	I	C	C

12. Assumptions, constraints and dependencies

Assumptions

- Users will tolerate a few pantry corrections a week if the app gives value first.
- A curated base of 300 regional dishes with variation rules covers most weeknight cooking in the launch region.
- Voice users speak about 4 minutes in a 15-minute cooking session.
- Households share one Kitchen Brain; individuals keep their own taste memories inside it.
- ₹88 per US$ for cost planning.

Constraints

- Stack fixed by the blueprint: React Native + Expo + TypeScript, Supabase, Claude-driven orchestration, Figma as design source, GitHub as code source.
- Real-time audio needs an Expo development build (not Expo Go).
- Entry Android phones (2–3 GB RAM) and 3G-class networks must be supported.
- No scraping or unofficial grocery integrations; no silent purchases.
- Personal data handled under India's DPDP Act 2023; voice audio is processed, not stored, by default.

External dependencies

Dependency	Used for	Fallback
Sarvam AI (speech-to-text, text-to-speech)	Indian-language voice	Gemini Live or another Indic speech provider behind the same voice interface
Anthropic Claude API (Haiku 5.5, Sonnet 5.5)	Conversation, planning, vision for scans	Models sit behind AIService; switchable
LiveKit (Cloud WebRTC + open-source Agents)	Real-time audio transport and turn-taking	Self-host LiveKit server
Supabase (Postgres, Auth, Storage, Edge Functions, Realtime, pgvector)	Kitchen Brain and backend	Postgres-compatible; portable
Open Food Facts	Barcode product lookup (V1.5)	Own product table grows from confirmed scans
Grocery providers (Zepto, Blinkit, BigBasket and others)	Carts and orders (V2)	Shopping list export
Meta WhatsApp Business Platform	Voice-note channel (V2)	In-app voice only
Expo Application Services (EAS), Sentry, PostHog	Builds, crash reports, analytics	Equivalents

13. Risks and mitigations

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-01	Pantry drifts out of sync and users stop trusting suggestions	High	High	Confidence score per item; automatic deduction after cooking; ask only questions that change tonight's plan	Backend / AI
R-02	Voice misunderstands dialects or kitchen noise	High	High	Indic-first speech model; on-device voice detection tuned for cooker whistles and frying; repeat-back confirmation; field testing in each dialect	Voice / AI engineer

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-03	AI invents unsafe or bad recipes	Medium	High	Curated base recipes plus variation rules; the model adapts within rules; food-safety checks (undercooked meat, egg-based mayo) in rescue and step logic	Culinary lead
R-04	Voice cost exceeds budget	Medium	High	The four cost controls in section 8; per-session cost telemetry from day one; free-tier caps	Backend / AI
R-05	Free tier economics don't close	High	High	Decide caps and partner funding before beta (section 8)	Product owner
R-06	Low-literacy users can't onboard	Medium	High	Voice-only onboarding, no login, field observation in pilot	Design
R-07	Privacy breach or consent failure on shared phones	Low	Very high	Row-level security on every table; no raw audio storage; voice-controlled "forget"; DPDP consent records	Privacy advisor
R-08	Grocery partners won't integrate officially	Medium	Medium	Shopping list works without partners; grocery agent stays V2	Product owner
R-09	Latency over 2 seconds on 3G	Medium	High	Streaming at every stage; India-region hosting; cached step audio; offline cook mode	Voice / AI engineer
R-10	Dependence on single speech vendor	Medium	Medium	Voice interface abstracts providers; keep a tested fallback	Voice / AI engineer
R-11	Scope creep delays the first magic moment	High	Medium	V1 scope fixed to section 5; every feature must use or improve the Kitchen Brain	Product owner

14. Approvals and sign-off

Role	Name	Decision	Date
Product owner / Founder, Drishix	Gaurav		
Engineering lead			
Design lead			

Role	Name	Decision	Date
Culinary lead			
Privacy advisor			

Open decisions to close at sign-off:

- [ ] Free-tier voice cap and funding model (section 8)
- [ ] Launch regional language (proposed: Rajasthani/Marwari-influenced Hindi plus one of Punjabi or Gujarati)
- [ ] Rural pilot partner (NGO or SHG federation)
- [ ] Final product name: Kitchnos vs Kitchen AI
- [ ] Sprint start date (proposed 16 Nov 2026)

15. Sources

- Dolmio/OnePoll: adults spend 37 hours a year choosing meals
- V-Guard/Momspresso survey of 500 Indian mothers (2020)
- ICMR-NIN Dietary Guidelines for Indians 2024 (South First)
- CEEW findings on Indian household diets (CSR Journal)
- UNEP Food Waste Index 2024 on India (NewsMeter)
- Spice Spoon: Andaza, cooking by feel
- NSO CMS-T 2025 on rural women's phone ownership (Business Standard)
- Quick commerce crosses 9 million daily orders (NewsBytes)
- The Spoon: Whirlpool lays off Yummly team
- Engadget: Samsung Food launch
- Plan to Eat: does a digital pantry inventory help?
- Sarvam AI: Speech to Text and API pricing
- LiveKit Cloud plans and billing docs
- Digital Applied: Claude Haiku 5.5 pricing
- eesel: Claude Sonnet 5.5 pricing
- Kitchen AI Product Blueprint and Kitchnos Adaptive Cooking Wireframe v4 (internal)